

Yining (Carina) Fu

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Professional Summary

Data-driven and strategic problem-solver with a Master's in Biostatistics and a proven track record of processing large-scale, complex datasets to drive actionable insights. Adept at leveraging advanced analytics (SQL, R, SAS) and data visualization to translate deep technical modeling into clear business stories for cross-functional stakeholders. Strong foundation in healthcare analytics ecosystem, encompassing clinical trial evaluation, spatial epidemiology, and understanding of medical billing structures (e.g., ICD-10, CPT) to inform strategic decision-making in healthcare.

Education

Emory University

Master of Public Health in Biostatistics

Atlanta, GA

Aug. 2024 – May. 2026

Relevant Coursework: Biostatistics, Regression Analysis, Epidemiologic Methods, Machine Learning, Clinical Trials, Statistical Computing, Categorical Data Analysis

North Carolina State University

Bachelor of Science in Statistics; Minor in Economics

Raleigh, NC

Aug. 2019 – Dec. 2023

Honors: Dean's Honors List, 4 semesters

Experience

Emory University

Quantitative Research Assistant

Atlanta, GA

Sep. 2025 – May 2026

- Integrated participant-level geospatial data with Census tract socioeconomic and environmental datasets to construct multidimensional neighborhood profiles; conducted data quality control and variable preprocessing, including transformation and standardization of ACS, PM2.5, population density, and built-environment indicators, successfully organizing and harmonizing a dataset of nearly 100,000 records to generate analysis-ready variables for downstream statistical modeling and neighborhood risk assessment.
- Applied and evaluated Self-Organizing Map (SOM) clustering to identify multidimensional neighborhood risk profiles, leveraging machine learning approaches beyond traditional statistical frameworks; used elbow, silhouette, and gap statistic methods to optimize cluster selection, resulting in an efficient standardized three-cluster solution that substantially improved between-cluster variance explanation compared with the unscaled approach (24.13% vs. 0.31%).
- Interpreted SOM-derived neighborhood clusters to characterize advantaged, moderate-risk urban, and concentrated-disadvantage profiles across socioeconomic and environmental dimensions, revealing complex neighborhood risk patterns in which higher walkability did not necessarily correspond to lower overall risk.

Chinese Academy of Sciences, Institute of Soil Science

Research Assistant

Nanjing, China

Jan. 2024 – Jun. 2024

- Developed spatial segmentation and contamination clustering analyses using SAS, R, and ArcGIS to identify regional heavy metal distribution patterns and stratify high-risk farmland areas for environmental monitoring.
- Performed environmental data preprocessing, normalization, and exploratory statistical analysis on soil physicochemical and heavy metal datasets, applying logarithmic transformation and spatial interpolation techniques to generate analysis-ready data for geospatial modeling.
- Interpreted regional soil contamination trends to support environmental risk assessment and remediation planning. Achieved the results of rehabilitating the farmland with excessive heavy metals by selecting a soil remediation scheme with reasonable soil passivation technology.

Academic Projects

Aedes-Borne Viruses (ABV) Transmission Analysis Project

R

- Analyzed data from a large-scale cluster-randomized trial using a Test-Negative Design to evaluate the efficacy of Targeted Indoor Residual Spraying against Dengue transmission.
- Developed custom R algorithms to calculate spatial clustering statistics, quantifying localized “neighborhood risk” against citywide background transmission.

- Leveraged ggplot2 to visualize complex spatial-temporal disease clusters, delivering clear, data-driven evidence to optimize resource allocation and targeted outbreak prevention strategies.

Clinical Trials Case Study

SAS

- Independently mined and processed large-scale clinical datasets using complex SAS PROC SQL queries, ensuring data integrity while tracking patient-level longitudinal outcomes.
- Engineered data architectures to derive critical clinical metrics, including Kaplan-Meier survival curves for time-to-first-event analysis and Kappa statistics to assess inter-rater reliability.
- Evaluated prognostic biomarker performance by calculating Sensitivity, Specificity, PPV, and NPV, translating complex diagnostic accuracy into structured evidence for potential preventative trial designs.

FAERS Machine Learning Project

R

- Cleaned and merged seven FDA Adverse Event Reporting System datasets to build an analysis-ready dataset for machine learning applications.
- Developed and compared LDA, QDA, and KNN classification models to predict adverse event outcomes, achieving 80% prediction accuracy.
- Evaluated model performance and conducted gap analysis to identify limitations and potential areas for improvement.

Prospective Evaluation of Radial Keratotomy (PERK) Group Project

R / SAS

- Conducted t-tests and ANOVA to assess relationships between patient characteristics and surgical outcomes.
- Produced numerical and graphical summaries to report post-surgical outcome improvements and procedure effectiveness.

Skills

Programming & Data Tools: R, Python, SAS, MySQL, Java

Statistical Methods: Regression Analysis, Hypothesis Testing, ANOVA, t-tests, Missing Data Imputation, Classification, Clustering, A/B Testing, Survival Analysis (Kaplan-Meier), Clinical trial

Packages & Platforms: Pandas, NumPy, ggplot2, dplyr, SAS Macro, SAS PROC SQL, Tableau, Microsoft Excel, PowerPoint

Languages: Mandarin Chinese, English